

Thesis Proposal: A Reduced-Order Aerodynamic State Modeling Framework for Real-Time Prediction and Control Using Sparse Sensors

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1 Motivation

Accurate aerodynamic characterization of bluff bodies—such as cylinders, simplified mannequins, and human athletes—is essential for performance optimization and control. Computational Fluid Dynamics (CFD) provides high-fidelity predictions for fixed geometries and controlled conditions, yet it remains computationally prohibitive for real-time applications, dynamic postures, and feedback-driven control. Furthermore, CFD cannot be directly integrated with noisy experimental sensor data in operational settings.

To address these gaps, this thesis proposes a unified, physics-informed Reduced-Order Modeling (ROM) and data-assimilation framework capable of predicting aerodynamic forces in real time using sparse pressure measurements. The goal is to develop a systematic modeling architecture applicable across a sequence of increasingly complex cases, from canonical flows to human posture aerodynamics.

2 Research Question

Can a coherent-structure-based reduced-order aerodynamic state model, equipped with a dynamic predictor and sensor-based corrector, be constructed to estimate and predict aerodynamic forces of bluff bodies in real time, and can this framework be extended across cases of increasing geometric and flow complexity?

3 Hypotheses

- **H1 (Coherent Structures):** Bluff-body wakes exhibit dominant coherent structures that can be captured in a low-dimensional modal or latent space.

- **H2 (Reduced Dynamics):** The temporal evolution of these structures can be represented by a reduced-order dynamic system preserving the relationship between state and aerodynamic forces.
- **H3 (Sensor-State Mapping):** Sparse pressure sensors contain sufficient information to estimate the reduced aerodynamic state and reconstruct drag in real time.
- **H4 (Framework Extensibility):** A common modeling architecture—state representation, predictor, and corrector—can be adapted to progressively complex cases with minimal structural modifications.

4 Objectives

Primary Objectives

1. Develop a reduced aerodynamic state representation using modal decomposition (POD/SPOD) or nonlinear latent embeddings.
2. Construct a dynamic predictor (e.g., Galerkin ROM, operator inference, Koopman model, or neural ODE) governing state evolution.
3. Design a measurement operator and data-assimilation corrector (EKF/EnKF/variational methods) to estimate aerodynamic state from sparse sensors.
4. Develop a data-driven state map (clustering + transition model) as an alternative predictor for drag reconstruction.
5. Validate extensibility across three stages: cylinder, simplified 3D dummy, and human body with limited degrees of freedom.

5 Methodology

5.1 Phase 1: Cylinder Flow (Proof of Concept)

- Acquire experimental data for three flow velocities using four pressure sensors.
- Extract coherent structures (POD/SPOD) and construct low-dimensional states.
- Train ROM and clustering-based state-transition models.
- Evaluate drag reconstruction accuracy and cross-velocity generalization.

5.2 Phase 2: Simplified 3D Dummy

- Instrument dummy with sparse sensors at aerodynamically relevant locations.
- Construct geometry-specific state representation and dynamic model.
- Validate relationships between posture variation, state signatures, and aerodynamic forces.

5.3 Phase 3: Human Body with Restricted Posture Space

- Define a limited set of posture degrees of freedom.
- Build a reduced-order predictive model for drag and lateral forces.
- Demonstrate potential for posture optimisation toward reduced drag.

6 Expected Contributions

- A general aerodynamic state modeling framework based on coherent structures.
- A real-time predictor–corrector system for estimating drag using sparse sensors.
- A data-driven aerodynamic state map applicable as an alternative to physics-based ROM.
- Multi-level validation from canonical to biomechanically relevant scenarios.
- Foundations for real-time aerodynamic optimisation for sports and human-in-the-loop control.

7 Timeline

1. Year 1:
 - Cylinder experiments, ROM design, sensor-state mapping.
 - Dummy experiments, model extension, data assimilation.
2. Year 3: Human posture modelling, optimisation layer, thesis consolidation.